

Omkar Oak

Tokyo, Japan

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[Linkedin](#) ◇ [Google Scholar](#) ◇ [GitHub](#)

Research Interests

Vision Language Models, Multimodal Large Language Models (MLLMs), Computer Vision, Natural Language Processing, and Deep Learning

Education

The University of Tokyo

MS, Information and Communication Engineering

Tokyo, Japan

Oct 2026 - present

- Advisor: [Prof. Toshihiko Yamasaki](#)
- Recieved the Japanese Government's MEXT Scholarship

College of Engineering Pune (COEP)

Bachelor of Technology in Computer Engineering

Pune, India

Aug 2021 - May 2025

- GPA: 9.52/10, Silver Medalist
- Advisor: [Prof. Suraj Sawant](#)
- Recieved the Frank Hirschgovel Academic Excellence Award

Experience

CVM Lab, The University of Tokyo

Researcher

Tokyo, Japan

Apr 2026 - present

- Advisor: Prof. Toshihiko Yamasaki
- Working on Vision Language Models and Multimodal LLMs for Disaster damage assessment using satellite imagery.

Wells Fargo Inc.

Software Engineer

Bangalore, India

Aug 2025 - Apr 2026

- Designed a ML workflow for fraud transaction detection in the Instant Payments team, using LangGraph based orchestrator and validator agents and autoencoder based anomaly detection models, including LSTM autoencoders and VAEs, to identify suspicious transactions across FedNow and RTP flows.
- Implemented natural language to MongoDB retrieval through a contextual intent builder, enabling operations teams to query payment and transaction data more effectively for fraud review and anomaly analysis.
- Built an advanced RAG pipeline with vector search and MMR, and added governance controls such as access control, PII masking, query safety checks, and LLM guardrails to support secure reporting, monitoring, and insight generation on large payment datasets.

COEP AI Lab

Research Assistant

Pune, India

Jul 2024 - May 2025

- Conducted research under Prof. Sawant, focusing on Computer Vision and Remote Sensing applications. Worked on image processing, semantic segmentation, change detection and VLMs.

Publications

6. [Omkar Oak](#), Rukmini Nazre, Rujuta Budke, Suraj Sawant, "TERRA-CD: A Multi-Temporal Benchmark Dataset for Urban Multi-Class and Semantic Change Detection" **ICICT** 2026. [[pdf](#)]
5. [Omkar Oak](#), Rukmini Nazre, Soham Naigaonkar Suraj Sawant, Amit Joshi, "A Novel Transfer Learning based CNN Model for Wildfire Susceptibility Prediction." **IEEE INCET** 2024. [[pdf](#)]
4. [Omkar Oak](#), Rukmini Nazre, Soham Naigaonkar, Suraj Sawant, Himadri Vaidya, "A Comparative Analysis of CNN-based Deep Learning Models for Landslide Detection." **IEEE ACOIT** 2024. [[pdf](#)]

3. Omkar Oak, Rukmini Nazre, Rujuta Budke, Yogita Mahatekar, "A Novel Multivariate Bi-LSTM model for Short-Term Equity Price Forecasting" **IEEE GCAT** 2024. [[pdf](#)]
2. Rukmini Nazre, Rujuta Budke, Omkar Oak, Amit Joshi, Suraj Sawant, "A Temporal Convolutional Network-based Approach for Network Intrusion Detection" **ICIICS** 2024. [[pdf](#)]
1. Omkar Oak, Rukmini Nazre, Rujuta Budke, Suraj Sawant, Rajkumar Saini, "DFANet: A Difference Fusion Attention-based method for Semantic Change Detection" **under review** . [[pdf](#)]

Projects

Receipt OCR using Fine-tuned VLMs, Vision Language Models [[github](#)]

Fine tuned SmolVLM using TRL with an SFT config based training pipeline, implementing both adapter-based (LoRA) and full fine-tuning variants on the SROIE v2 dataset, achieving a test CER of 0.313 compared to SROIE box texts.

Financial Report Summarizer, Data Mining, NLP [[github](#)]

Developed an extractive summarizer for SEC 10K filings with 85% accuracy and 50% speed improvement, including a complete end-to-end ML pipeline with web scraping and information retrieval using NLP techniques.

Time Series Analysis of Forex Rates, Machine Learning [[github](#)]

Time series analysis and forecasting using ARIMA and LSTM models for Forex rates of 37 currencies to INR for 2006-2023, including TSA preprocessing, trend/seasonality analysis, stationarity conversion, and ACF calculations.

Priority Scheduling in xv6, Operating Systems [[github](#)]

Implemented priority scheduling in xv6 using C and assembly language for kernel modifications, including priority queues that allow dynamic adjustment of process priorities based on predefined criteria.

Awards and Scholarships

- Recieved the Japanese Government's **MEXT Scholarship** for a fully-funded Master's at the University of Tokyo
- Awarded the **Silver Medal** (Rank: 2/178) in the Department of Computer Engineering at COEP
- Received the **Frank Hirschvogel Award** for Outstanding Graduation results from Hirschvogel Stiftung, Germany
- **Won** the **Best Technical Implementation Award** for Bachelor's Thesis
- **Secured Rank 1** in the Department of Computer Engg. in Sem V, Sem VI and Sem VIII
- Passed the **Japanese Language Proficiency Test (JLPT) N3** exam conducted by the Embassy of Japan
- **Secured Rank 70** out of 400,000 students scoring 99.93 percentile in MHT-CET (Engineering Entrance Exam)

Skills

Languages: C, C++, Java, Python, R, Julia, Javascript, SQL, Bash

Technologies: Git, Docker, Kubernetes, AWS, Linux, Hadoop

Libraries: TensorFlow, Keras, NumPy, Pandas, Scikit-learn, PyTorch, OpenCV

Computing Coursework: Operating Systems, Computer Networks, Database Systems, Data Structures, Web Systems, Artificial Intelligence, Data Science, Microprocessors, Computer Organization, Theory of Computation, Analysis of Algorithms, Compilers, Cloud Computing, Geographic Information Systems

Math Coursework: Linear Algebra, Probability and Statistics, Vector Calculus, Partial Differential Equations, Multivariate Calculus, Discrete Structures and Graph Theory

Interdisciplinary Coursework: Optics and Modern Physics, Robotics, Sensors and Automation

Languages and Test Scores

English: TOEFL 118/120 **Japanese:** JLPT N3 **GRE:** 321/340 (Q: 168, V: 153)